THE ADIC* PROPERTY DESCENDS ALONG A TAUT MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a chain of containments and one identification of a quotient of the target by the extension of the chosen ideal; both lie inside the counted span and are reproduced verbatim.

Editorial note. This candidate had been set aside earlier because the shared gate reported a public-item collision; after the gate was repaired the collision disappeared and the candidate was re-admitted, the re-run being recorded in the queue.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Weak adicness and admissibility of the source from the corresponding descent statement. Same chapter, Lemma lemma-taut-descent-weakly-adic, invoked at proof lines 1899-1901. Reduces to showing that a suitable ideal of definition has open power closures.
- The closure of an ideal is the intersection of its images, and an ideal consisting of topologically nilpotent elements lies in the Jacobson radical. Same chapter, Lemma lemma-closed at proof lines 1909-1911 and Lemma lemma-weakly-admissible-henselian at proof lines 1917-1919. Produce the containment modulo the square of the ideal and place the ideal in the Jacobson radical.
- Nakayama's lemma, part (2). Algebra, Lemma algebra-lemma-NAK part (2), invoked at proof lines 1919-1921. Forces the finitely generated quotient module to vanish.
- An inclusion of ideals can be tested after a faithfully flat extension. Standard, used at proof lines 1930-1936. Descends the containment of powers to the source.
- The criterion for a weakly adic ring with a finitely generated ideal of definition to be adic. Same chapter, Lemma lemma-weakly-adic-finite-generation-bis, invoked at proof lines 1946-1948. Gives the conclusion.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AMS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 922–922. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1857–1857. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1633–1633. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1699–1699. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0DCZ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1116–1116. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *In the situation above, if B is adic and has a finitely generated ideal of definition and A is complete, then A is adic and has a finitely generated ideal of definition.*

Proof. We already know that A is weakly adic and a fortiori admissible by Lemma 0GXP (and Lemma 0GXF to see that adic rings are weakly adic). Let $I \subset A$ be an ideal of definition. Let $J \subset B$ be a finitely generated ideal of definition. Since the closure of IB is open, we can find an $n > 0$ such that J^n is contained in the closure of IB. Thus after replacing J by J^n we may assume J is a finitely generated ideal of definition contained in the closure of IB. By Lemma 0AMS this certainly implies that

$$J \subset IB + J^2$$

Consider the finitely generated A-module $M = (J + IB)/IB$. The displayed equation shows that $JM = M$. By Lemma 0DCZ (for example) we see that J is contained in the Jacobson radical of B. Hence by Nakayama's lemma, more precisely part (2) of Algebra, Lemma 00DV, we conclude $M = 0$. Thus $J \subset IB$.

Since J is finitely generated, we can find a finitely generated ideal $I' \subset I$ such that $J \subset I'B$. Since $A \to B$ is continuous, $J \subset B$ is open, and I is an ideal of definition, we can find an $n > 0$ such that $I^n B \subset J$. Let $J_{n+1} \subset B$ be the closure of $I^{n+1}B$. We have

$$I^n \cdot (B/J_{n+1}) \subset J \cdot (B/J_{n+1}) \subset I' \cdot (B/J_{n+1})$$

Since $A/I^{n+1} \to B/J_{n+1}$ is faithfully flat, this implies $I^n \cdot (A/I^{n+1}) \subset I' \cdot (A/I^{n+1})$ which in turn means

$$I^n \subset I' + I^{n+1}$$

This implies $I^n \subset I' + I^{n+k}$ for all $k \geq 1$ which in turn implies that $I^{nm} \subset (I')^m + I^{nm+k}$ for all $k, m \geq 1$. This implies that the closure of $(I')^m$ contains I^{nm}. Since the closure of I^{nm} is open as A is weakly adic, we conclude that the closure $(I')^m$ is open for all m. Since these closures form a fundamental system of open ideals of A (as the same thing is true for the closures of I^n) we conclude by Lemma 0GXG. □